Practice Set 21 SAQ Sheet – Short Answer Practice

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Short-answer practice sheet compiled from handwritten study lists.

Q3a. Cystinuria

Type: SAQ (10 marks)

Definition and genetics

- **Autosomal recessive** defect of proximal tubular (and jejunal) reabsorption of the dibasic amino acids **COLA: cystine, ornithine, lysine, arginine**. Only cystine is insoluble, so only cystine forms stones.
- Transporter subunits and ICC (International Cystinuria Consortium) types:

Gene	Subunit	Chromosome	Type
SLC3A1	rBAT (heavy)	2p	A
SLC7A9	$b^{0,+}$AT (light)	19q	B
Both	—	2 and 19	AB

- Fractional excretion of cystine rises from about 0.4% to up to 100%. Incidence 1 in 1000 to 1 in 17,000; **1–3% of adult stones, up to 10% of paediatric stones**; males more severe.

Pathophysiology

- **Pure supersaturation disease**: no specific inhibitor. Solubility is pH dependent: about 300 mg/L at pH 5, 400 mg/L at pH 7, 1000 mg/L at pH 9.
- Coexisting hypocitraturia (commonest), hyperuricosuria or hypercalciuria cause mixed calcium–cystine stones.

Clinical features

- First stone at mean about 12 years; recurrent, bilateral, staghorn stones; CKD (chronic kidney disease) from repeated surgery (about a quarter of adults have eGFR [estimated glomerular filtration rate] <60).

Diagnosis

- **Hexagonal crystals** in urine (always abnormal); cyanide–nitroprusside screen; 24-hour urinary cystine (normal <60 mg/day/1.73 m^2; homozygotes often >400).
- Stone: yellow, waxy, faintly radiopaque (sulphur); **resists SWL (shock wave lithotripsy)**.

Management

- **Fluids:** urine output **≥3 L/day**, including at night.
- **Diet:** sodium ≤2300 mg/day; limit animal protein (not in children).
- **Alkalinise to pH 7.0–7.5** with potassium citrate 30–60 mEq/day; avoid >7.5 (calcium phosphate stones) and sodium alkali (raises cystine excretion).
- **Thiols** if cystine stays >250 mg/L: thiol–cystine mixed disulfide is soluble.
  - **Tiopronin** (choice): 300 mg twice daily, up to 1200 mg/day.
  - **D-penicillamine:** 250 mg/day, titrated; rash, proteinuria, nephrotic syndrome, pancytopenia.
  - Captopril: weak evidence.
- Monitor blood count, liver tests, urine protein.
- **Surgery:** ureteroscopy or PCNL (percutaneous nephrolithotomy) with laser; nephron-sparing approach.

Diagram to draw: proximal tubule cell with rBAT/$b^{0,+}$AT transporter and hexagonal crystal.

Key points to remember:

- Autosomal recessive; SLC3A1 (type A), SLC7A9 (type B).
- Hexagonal crystals; target cystine <250 mg/L.
- Urine ≥3 L, pH 7.0–7.5, then tiopronin.

Q3b. TURP techniques

Type: SAQ (10 marks) (irrigation fluids answered in 2017-Summer Paper 2, Q3a; complications in Practice Set 9 Paper 2, Q3b)

Introduction

- TURP (transurethral resection of the prostate): wire loop removes transition-zone adenoma **from bladder neck to verumontanum, down to the surgical capsule**. Gold standard for glands up to 80 mL (AUA [American Urological Association] 2023); larger glands need enucleation or simple prostatectomy.

Preparation

- Sterile urine; spinal or general anaesthesia; lithotomy with peroneal padding.
- Irrigant at body temperature, lowest height giving vision.
- **Inspect bladder first** (tumour, diverticula); identify ureteric orifices, verumontanum, external sphincter.

Resection techniques

- **Median lobe first**, then lateral lobes.
- **Modified Nesbit "encirclement":** start at 11–9 and 1–3 o'clock; expose but do not resect bladder neck fibres.
- **Floor-first:** channel at 5 or 7 o'clock to the capsule (sets depth early); widen laterally; lateral lobes "fall" into the fossa.
- **Anterior start:** 12 o'clock bladder neck, then quadrants down to the apex.
- **Chips:** long "canoe-like" chips with synchronised rocking.
- **Anterior tissue and apex last** in a bloodless field; leaving a little tissue is better than incontinence.
- Haemostasis: arterial bleeders must be fulgurated; venous ooze settles with catheter traction.
- Evacuate chips (Ellik evacuator); three-way catheter.

Energy variants

	Monopolar	Bipolar
Current	Loop → patient → pad	Stays at loop
Irrigant	Glycine, sorbitol, water	Saline
TUR (transurethral resection) syndrome	Yes; keep resection <90 min	Eliminated
Bleeding	More	Less transfusion, clot retention
Efficacy	Equal	Equal

- Plasma loop (platinum–iridium) cuts at lower temperature and seals vessels.
- **Also know:** TUVP (transurethral vaporisation, button electrode; ≤80 mL); TUIP (transurethral incision, <30 mL, no median lobe); bipolar enucleation; high bleeding risk: AUA prefers HoLEP (holmium laser enucleation) or PVP (photoselective vaporisation).

Outcomes

- About 80% symptom improvement; low retreatment; retrograde ejaculation about two-thirds.

Diagram to draw: resection field with clock positions, bladder neck, verumontanum and sphincter.

> **Key points to remember:**
>
> - Bladder neck to verumontanum, depth to capsule.
> - Median lobe first; apex last.
> - Bipolar saline TURP: no TUR syndrome.

Q3c. Amyloidosis in urology

Type: SAQ (10 marks)

Definition

- Extracellular deposits of misfolded protein in **β-pleated sheet fibrils**. **Congo red stain gives apple-green birefringence** under polarised light.

Classification

Type	Protein	Cause
AL (amyloid light-chain; primary)	Immunoglobulin light chain	Plasma cell dyscrasia, myeloma
AA (amyloid A; secondary)	Serum amyloid A	Chronic inflammation: TB (tuberculosis), rheumatoid arthritis, osteomyelitis, pressure sores, hidradenitis
ATTR (hereditary transthyretin)	Transthyretin	Autosomal dominant neuropathy
Dialysis-related	β2-microglobulin	Long-term dialysis
Localised	Usually AL	Deposits in one organ only

Urological sites

- **Kidney (systemic):** nephrotic proteinuria, CKD (chronic kidney disease), enlarged kidneys, renal vein thrombosis, proximal tubular defects.
- **Bladder (usually localised): painless gross haematuria**; yellowish plaques or nodules that **mimic carcinoma**. Secondary bladder amyloid can bleed severely; Campbell lists amyloidosis among systemic causes of haemorrhagic cystitis.
- **Ureter and renal pelvis:** stricture or filling defect, often calcified; mimics urothelial cancer; obstruction.
- **Urethra:** rare; mass, obstructive voiding, haematuria.
- **Seminal vesicles:** senile deposits in older men; can mimic cancer invasion on MRI (magnetic resonance imaging).
- **Also know:** ATTR gives neurogenic bladder (underactive detrusor, LUTS [lower urinary tract symptoms] >80%) and erectile dysfunction; adrenal insufficiency; papillary necrosis.

Evaluation

- **Biopsy is diagnostic** (avoids unnecessary nephroureterectomy or cystectomy).
- Exclude systemic disease: serum and urine protein electrophoresis, immunofixation, free light chains, echocardiogram, renal function.

Management

- **Localised bladder:** TUR (transurethral resection) or fulguration; intravesical DMSO (dimethyl sulfoxide) reported; recurrence is common, so follow-up cystoscopy is needed.
- **Ureter:** ureteroscopic resection or segmental excision; stent for obstruction.
- **Urethra:** observe if mild; dilatation or urethroplasty if obstructive.
- **Refractory bleeding:** irrigation, embolisation, cystectomy.
- **Systemic:** AL: chemotherapy (daratumumab with bortezomib regimen, ANDROMEDA trial) or stem cell transplant; AA: treat inflammation (colchicine in familial Mediterranean fever); ATTR: tafamidis, patisiran. AA amyloid can recur in a renal transplant.

Diagram to draw: Congo red stained fibrils with apple-green birefringence.

Key points to remember:

- Congo red, apple-green birefringence.
- Localised bladder amyloid mimics cancer; biopsy then TUR.
- Always exclude systemic AL disease.

Q3d. Management of CAPD peritonitis

Type: SAQ (10 marks) (CAPD as a whole answered in 2018-Summer Paper 2, Q3e and Practice Set 1 Paper 2, Q3a)

Introduction

- CAPD (continuous ambulatory peritoneal dialysis) peritonitis is the leading cause of technique failure and transfer to haemodialysis. Commonest organism: **coagulase-negative *Staphylococcus*** (touch contamination).
- ISPD (International Society for Peritoneal Dialysis) 2022 target: centre rate **≤0.40 episodes per patient-year**.

Diagnosis (any 2 of 3)

1. Abdominal pain and/or **cloudy effluent**.
2. Effluent **WCC (white cell count) >100/µL with >50% neutrophils** (dwell ≥2 hours).
3. Positive effluent culture.

Initial steps

- Effluent cell count, Gram stain, **culture in blood-culture bottles** (culture-negative rate should be <15%).
- Examine exit site and tunnel; analgesia; heparin in bags if fibrin.

Empirical antibiotics

- **Intraperitoneal route preferred** (intravenous if septic); start immediately after samples.
- Cover **Gram-positives** (vancomycin or first-generation cephalosporin, e.g. cefazolin) **and Gram-negatives** (third-generation cephalosporin, e.g. ceftazidime, or aminoglycoside). Choice follows centre sensitivities.
- Intermittent (once daily in long dwell) or continuous dosing; avoid prolonged aminoglycoside (residual function, ototoxicity).
- **Antifungal prophylaxis** (nystatin or fluconazole) during antibiotics.

Organism-specific duration

Organism	Duration
Coagulase-negative *Staphylococcus*, streptococci, culture-negative	2 weeks
S. aureus, enterococcus, Gram-negatives	3 weeks
Pseudomonas	3 weeks, two antipseudomonal drugs
Polymicrobial enteric	Add metronidazole; surgical review (bowel perforation)
Fungal	**Immediate catheter removal** + antifungal

Catheter removal

- **Refractory** (effluent not clear after 5 days of right antibiotics), relapsing, fungal, refractory exit-site or tunnel infection.
- Haemodialysis meanwhile; new catheter after at least 2 weeks.

Definitions

- **Relapse:** same organism (or sterile) within 4 weeks of finishing therapy. **Recurrent:** different organism within 4 weeks. **Repeat:** same organism after 4 weeks.

Prevention

- Training, exit-site mupirocin or gentamicin, prophylaxis before colonoscopy or gynaecological procedures.

Diagram to draw: management flowchart from cloudy bag to catheter removal.

> **Key points to remember:**
> - Two of: cloudy fluid or pain, WCC >100/μL with >50% neutrophils, positive culture.
> - Intraperitoneal Gram-positive plus Gram-negative cover.
> - Remove catheter for fungal or refractory (5 days) peritonitis.

Q3e. Avanafil

Type: SAQ (10 marks)

Introduction

- Second-generation oral **PDE5I (phosphodiesterase type 5 inhibitor)**; **FDA (US Food and Drug Administration) approved 2012**; Europe 2013. Structure differs most from the other PDE5Is.

Mechanism

- Sexual stimulation releases NO (nitric oxide), which raises cGMP (cyclic guanosine monophosphate) and relaxes cavernosal smooth muscle. PDE5 breaks down cGMP; avanafil blocks this.
- **Enhances, does not initiate, erection**: stimulation is needed.
- **More selective for PDE5** over PDE6 (retina) and PDE1 than sildenafil, so **no visual disturbance** in the comparison table.

Pharmacokinetics (Campbell Table 72.7)

Feature	Avanafil	Sildenafil	Tadalafil
Time to peak	**0.3–0.5 h (fastest)**	0.8 h	2 h
Half-life	3–5 h	3–5 h	17.5 h
Fatty food	Reduces absorption	Reduces	No effect
Doses	**50, 100, 200 mg**	25–100 mg	5–20 mg
Blue vision	No	Yes	Rare

- Taken about 15–30 minutes before sex; start 100 mg; maximum once daily. Metabolised by CYP3A4 (cytochrome P450 3A4).

Efficacy

- Effective in general ED (erectile dysfunction), **diabetes** and **after nerve-sparing radical prostatectomy** (REVIVE trial programme): improves IIEF (International Index of Erectile Function) erectile scores and successful-intercourse rates, some within 15 minutes. Efficacy equivalent across PDE5Is; no head-to-head superiority.

Adverse effects

- Headache, flushing, nasal congestion, dyspepsia; back pain rare.

Contraindications and interactions

- **Nitrates in any form**: absolute; no nitrate within 12 hours of a dose.
- Avoid strong CYP3A4 inhibitors (ketoconazole, ritonavir); maximum 50 mg with moderate inhibitors.
- **α-blockers:** start 50 mg once stable (additive hypotension).
- Caution: aortic stenosis, recent myocardial infarction or stroke, hypotension, retinitis pigmentosa. Not recommended in severe renal impairment, dialysis or severe hepatic impairment.
- **Also know:** proposed prophylaxis for recurrent ischaemic priapism; can cause priapism rarely.

Advantages

- **Fastest onset**, fewer visual and muscle effects; suits on-demand spontaneous use.

Diagram to draw: NO–cGMP pathway in cavernosal smooth muscle with the PDE5 block.

> **Key points to remember:**
>
> - Peak 0.3–0.5 h; doses 50, 100, 200 mg.
> - Highly PDE5-selective: no blue vision.
> - Nitrates absolutely contraindicated.

Q3f. Methods of orchidopexy

Type: SAQ (10 marks) (treatment algorithm of undescended testis answered in 2018-Summer Paper 3, Q3a)

Principles

- Operate **after 6 months, ideally before 18 months** (Kollin RCT [randomised controlled trial]: surgery at 9 months gave better testicular growth than at 3 years). No hormones.
- Goals: tension-free **dependent scrotal** testis with intact blood supply; ligate the patent processus.

Palpable testis

- **Inguinal orchidopexy:** groin incision → open external oblique (spare ilioinguinal nerve) → free cord, divide gubernaculum → separate and **ligate sac at internal ring** → retroperitoneal mobilisation, divide lateral bands → **Prentiss manoeuvre** (medial transposition under inferior epigastric vessels) → **subdartos pouch**. Transalbugineal sutures unnecessary.
- **Trans-scrotal (Bianchi) orchidopexy:** high scrotal incision; sac ligated through it; inguinal conversion 3.5–4.4%; atrophy 0.3–0.6%; less pain.

Non-palpable testis: examination under anaesthesia, then laparoscopy

Method	Principle	Success
Primary laparoscopic	Mobilise vessels, preserve vasal collaterals, new hiatus medial to epigastric vessels	>90% (97%)
Open transabdominal (Jones and Bagley)	Preperitoneal high mobilisation	95%
Fowler–Stephens (one- or two-stage)	Divide testicular vessels high; testis survives on **vasal and gubernacular collaterals**; second stage after 6 months	60–97% (one-stage 63%, two-stage 68%)
Shehata traction	Fix testis near opposite anterior superior iliac spine; vessels lengthen; orchidopexy at 12 weeks	84–85%; atrophy 10%
Microvascular autotransplantation	Testicular vessels to inferior epigastric vessels	96%
Orchidectomy	Postpubertal, dysgenetic or atrophic testis	—

- Fowler–Stephens for testes **not within 2–4 cm of the internal ring**.
- **Also know:** staged (two-stage) inguinal orchidopexy; Dessanti PTFE (polytetrafluoroethylene) sleeve traction; gubernaculum-sparing technique; redo approaches (Redman lateral, Cartwright fascial patch).

Complications

- **Atrophy** (highest after Fowler–Stephens), retraction or reascent, vas or vessel injury, haematoma, infection; torsion very rare after subdartos fixation.

Follow-up

- Size and position at 6 months; counsel on fertility and cancer risk; self-examination after puberty. Offer prosthesis for an atrophic testis.

Diagram to draw: Fowler–Stephens collateral supply: testicular artery clipped, deferential and cremasteric arteries intact.

Key points to remember:

- Orchidopexy at 6–18 months; subdartos pouch.
- Prentiss manoeuvre gives extra length.
- Fowler–Stephens relies on vasal collaterals.

Rapid Revision Sheet – Practice Set 21 SAQ Sheet

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

Q3a. Cystinuria (SAQ)

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- Hexagonal crystals; target cystine <250 mg/L.
- Urine ≥3 L, pH 7.0–7.5, then tiopronin.

Q3b. TURP techniques (SAQ)

- Bladder neck to verumontanum, depth to capsule.
- Median lobe first; apex last.
- Bipolar saline TURP: no TUR syndrome.

Q3c. Amyloidosis in urology (SAQ)

- Congo red, apple-green birefringence.
- Localised bladder amyloid mimics cancer; biopsy then TUR.
- Always exclude systemic AL disease.

Q3d. Management of CAPD peritonitis (SAQ)

- Two of: cloudy fluid or pain, WCC >100/μL with >50% neutrophils, positive culture.
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Q3f. Methods of orchidopexy (SAQ)

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